

A disproof of Conjecture 00000000030

Squares of a large set $A \subseteq [N]$ need not be intersective

1 The conjecture

Conjecture 00000000030 reads:

Sparsification of intersective sets: there exists an absolute $c > 0$ such that whenever $A \subset [N]$ with $|A| \geq N^{1/2-c}$, the set $\{a^2 : a \in A\}$ is intersective (it has nonempty intersection with every set of positive density).

The quantifiers are: $\exists c > 0 \forall N \forall A \subseteq [N]$ with $|A| \geq N^{1/2-c}$, the set $S_A = \{a^2 : a \in A\}$ is intersective. The statement is universal (“whenever”), and this universal reading is the one we refute. We show that it fails for *every* $c > 0$ and *every* $N \geq 3$, under the standard notion of intersective set and under the statement’s own parenthetical, with either convention $[N] = \{1, \dots, N\}$ or $[N] = \{0, \dots, N-1\}$. The reason is the classical local obstruction modulo 3.

2 Definitions

We use the conventions of Bienvenu, Griesmer, Le and Lê, *Intersective sets for sparse sets of integers* (arXiv:2401.07758, <https://arxiv.org/html/2401.07758v1>). For $E \subseteq \mathbb{N}$ write $|E \cap [N]|$ with $[N] = \{1, \dots, N\}$.

Definition 1. The *upper density* of $E \subseteq \mathbb{N}$ is $\bar{d}(E) = \limsup_{N \rightarrow \infty} |E \cap [N]|/N$. The set E has *natural density* d if $|E \cap [N]|/N \rightarrow d$.

Definition 2 (intersective set). A set $R \subseteq \mathbb{N}$ is *intersective* if every $E \subseteq \mathbb{N}$ with $\bar{d}(E) > 0$ contains two distinct elements $x \neq y$ with $x - y \in R$; equivalently $R \cap (E - E) \neq \emptyset$, the difference being nonzero.

The paper (Definition 1.1, with $E = \mathbb{N}$) says: “a subset $R \subset \mathbb{N}$ is said to be E -intersective if for every $A \subset E$ with $\bar{d}_E(A) > 0$, we have $R \cap (A - A) \neq \emptyset$. Thus, R is E -intersective if every A satisfying $\bar{d}_E(A) > 0$ contains two distinct elements differing by an element of R ... if $E = \mathbb{N}$, then R is simply called intersective”. It also recalls that Sarkozy and Furstenberg proved that $\{n^2 : n \in \mathbb{N}\}$ is intersective, and remarks: “An intersective set must contain a nonzero multiple of every natural number.” This remark is exactly the obstruction used below; we do not rely on it, but prove what we need.

We refute four notions at once (in Lean: `IsIntersective`, `IsIntersectiveNat`, `MeetsDenseSets`, `MeetsDenseSetsNat`):

- (1) Definition 2 (upper density);
- (2) the weaker requirement where only sets E of positive *natural* density are tested;

(3) the literal parenthetical: $R \cap E \neq \emptyset$ for every E with $\bar{d}(E) > 0$;

(4) the same with positive natural density.

In (1) and (2) the condition is formalized as $\exists x, y \in E, x \neq y \wedge \exists r \in R, r = x - y$ in $\mathbb{Z}$ (two distinct elements, so a set containing 0 is not trivially intersective). Since one fixed test set $E = 3\mathbb{N}$ has natural density $1/3$, it has positive density in every usual sense (natural, lower, upper, upper Banach), so the refutation does not depend on which density is meant.

3 The counterexample

Lemma 3. *For $m \geq 1$, $|m\mathbb{N} \cap [N]| = \lfloor N/m \rfloor$. Hence $m\mathbb{N}$ has natural density $1/m$ and upper density $1/m$.*

Proof. The count is Mathlib's `Nat.Ioc_filter_dvd_card_eq_div`. Since $(N - (m - 1))/m \leq \lfloor N/m \rfloor \leq N/m$, the ratio is squeezed between $1/m - \frac{m-1}{m} \cdot \frac{1}{N}$ and $1/m$, so it tends to $1/m$; a convergent sequence has limsup equal to its limit. $\square$

Lemma 4. *If $R \subseteq \mathbb{N}$ and every $r \in R$ satisfies $r \equiv 1 \pmod{3}$, then R is not intersective in any of the senses (1)–(4).*

Proof. Take $E = 3\mathbb{N}$, of natural density $1/3 > 0$ (Lemma 3). No $r \in R$ is a multiple of 3, so $R \cap E = \emptyset$. If $x, y \in E$ then $x - y \in 3\mathbb{Z}$, while $r \equiv 1 \pmod{3}$, so $r \neq x - y$; thus $R \cap (E - E) = \emptyset$. $\square$

For $N \geq 0$ let

$$A_N = \{a \in \{0, \dots, N-1\} : 3 \nmid a\}.$$

Then $0 \notin A_N$, so $A_N \subseteq \{1, \dots, N\}$ and $A_N \subseteq \{0, \dots, N-1\}$.

Lemma 5. $2N \leq 3|A_N| + 2$ for every N . Consequently, for $N \geq 3$ and every $c \geq 0$,

$$N^{1/2-c} \leq \sqrt{N} \leq |A_N|.$$

Proof. Each block $\{3t, 3t+1, 3t+2\}$ contributes two elements, which gives $2N \leq 3|A_N| + 2$ by induction in steps of 3 (base cases $N < 3$ checked directly). For $N \geq 3$ this gives $|A_N| \geq 2$ and $N \leq 2|A_N|$, so $N \leq 2|A_N| \leq |A_N|^2$ and $\sqrt{N} \leq |A_N|$. Finally $N \geq 1$ and $1/2 - c \leq 1/2$ give $N^{1/2-c} \leq N^{1/2} = \sqrt{N}$. $\square$

Theorem 6 (main). *For every $c \geq 0$ and every $N \geq 3$: $A_N \subseteq \{1, \dots, N\}$, $A_N \subseteq \{0, \dots, N-1\}$, $|A_N| \geq N^{1/2-c}$, $|A_N| \geq \sqrt{N}$, and $S_{A_N} = \{a^2 : a \in A_N\}$ is not intersective in any of the senses (1)–(4).*

Proof. If $3 \nmid a$ then $a \equiv \pm 1 \pmod{3}$ and $a^2 \equiv 1 \pmod{3}$. Apply Lemmas 5 and 4. $\square$

Corollary 7. *Conjecture 00000000030 is false, for each of the notions (1)–(4) and for both conventions for $[N]$. Explicitly: for any $c > 0$, $N = 3$ and $A = \{1, 2\}$ satisfy $|A| = 2 \geq 3^{1/2-c}$, but $\{1, 4\}$ misses both $3\mathbb{N}$ and $3\mathbb{N} - 3\mathbb{N}$.*

4 Other readings

- *Finitary reading.* If “intersective” is meant at scale M (every $B \subseteq [M]$ with $|B| \geq \delta M$ has two elements differing by some a^2 , $a \in A$), take $B_M = 3\mathbb{N} \cap [M]$: $|B_M| = \lfloor M/3 \rfloor \geq (M-2)/3$, and neither B_M nor $B_M - B_M$ contains a square of an element of A_N . This holds for every M and N (Lean: `finitary_witness`), so it refutes any threshold $\delta \leq 1/4$ for $M \geq 8$.
- *Infinite A .* For $A_\infty = \{a : 3 \nmid a\}$ one has $|A_\infty \cap [N]| \geq N^{1/2-c}$ for all $N \geq 3$, and $\{a^2 : a \in A_\infty\}$ is not intersective (Lean: `infinite_version`).
- *Any finite A .* No finite set of positive integers is intersective in senses (1)–(4): test it against $m\mathbb{N}$ with $m > \max R$ (Lean: `finite_not_intersective`, `no_finite_square_set_intersective`). This only concerns finite sets of positive integers and the infinitary notions (1)–(4). It does *not* refute an existential claim about a well-chosen infinite square set, nor an existential finitary (quantitative) sparsification statement; neither is what the conjecture says.

The mod-3 argument does not use finiteness: it rules out the universal claim even for sets A of positive density. The reading refuted in this note is the one written, namely the universal statement (“whenever $A \subset [N]$ with $|A| \geq N^{1/2-c}$ ”), in the senses (1)–(4) and in the finitary form above.

5 Lean 4 formalization

The directory `lean/` is a Lean 4 project (Lean v4.33.1, Mathlib v4.33.1); the source is `Conjecture30/Basic.lean` (namespace `C30`). It defines `countIn E N = |E ∩ {1, ..., N}|`, `upperDensity E` (a `limsup` in $\mathbb{R}$), `HasNatDensity E d` (a `Tendsto` statement), the four predicates above, and

`Conjecture box P :  $\iff \exists c > 0, \forall N \forall A, A \subseteq \text{box } N \rightarrow (N : \mathbb{R})^{1/2-c} \leq |A| \rightarrow P(\{a^2 : a \in A\})$ .`

Main results:

- `hasNatDensity_multiples`, `upperDensity_multiples` (Lemma 3);
- `not_intersective_of_mod3` (Lemma 4); `goodSet_size` (Lemma 5), with `goodSet N = A_N`;
- `main_family` (Theorem 6), for all $c \geq 0$ and $N \geq 3$;
- `conjecture_false`: the conjunction of $\neg \text{Conjecture } B \quad P$ for $B \in \{\text{Finset.Icc } 1, \text{Finset.range}\}$ and $P \in \{\text{IsIntersective}, \text{IsIntersectiveNat}, \text{MeetsDenseSets}, \text{MeetsDenseSetsNat}\}$;
- `finitary_witness`, `infinite_version`, `finite_not_intersective`, `no_finite_square_set_intersective` (Section 4).

There is no `sorry`, no `native_decide` and no added axiom. `#print axioms` for `C30.conjecture_false`, `C30.main_family`, `C30.finitary_witness`, `C30.infinite_version` and `C30.no_finite_square_set_intersective` reports only `propext`, `Classical.choice` and `Quot.sound`. The file also compiles with `-DwarningAsError=true`. To check:

```
cd lean
lake exe cache get
lake build
```